

Nishan Bhattarai, Ph.D.

Assistant Professor, Department of Geography & Environmental Sustainability
University of Oklahoma, Norman, OK 73019
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[Website](#) [Google Scholar](#), [ORCID](#), [ResearchGate](#)

ACADEMIC APPOINTMENTS

- 2022- *Assistant Professor*, Department of Geography & Environmental Sustainability, University of Oklahoma, Norman, OK
Affiliate, Data Institute for Societal Challenges (DISC)
Affiliate, Institute for Resilient Environmental and Energy Systems (IREES)
- 2020-2022 *Postdoctoral Research Associate (Research Physical Scientist)*, Hydrology and Remote Sensing Lab, USDA-ARS, Beltsville, MD/Supervisor: Dr. William P. Kustas
- 2016-2020 *Postdoctoral Research Fellow*, School for Environment & Sustainability, University of Michigan, Ann Arbor, MI/Supervisor: Dr. Meha Jain
- 2015-2016 *Postdoctoral Fellow*, Fletcher School of Law & Diplomacy, Tufts University, MA/Supervisor: Dr. Avery Cohn
- 2011-2014 *Teaching Assistant*, Dept. of Environmental Resources Engineering, State University of New York, College of Environmental Science & Forestry (SUNY-ESF), Syracuse, NY
- 2010-2011 *Research Project Assistant*, Dept. of Environmental Resources Engineering, SUNY-ESF, Syracuse, NY
- 2008-2010 *Research Assistant*, Department of Biosystems Engineering, Auburn University, AL

EDUCATION

- Ph.D. SUNY College of Environmental Science & Forestry, Syracuse, NY
Department of Environmental Resources Engineering
Dissertation Title: *Single-source surface energy balance algorithms to estimate evapotranspiration from satellite-based remotely sensed data*
Supervisor: Dr. Lindi J. Quackenbush
- MS Auburn University, Auburn AL
Department of Biosystems Engineering and School of Forestry & Wildlife Sciences
Dissertation Title: *Use of Remotely Sensed Data to Quantify Plant Water Use from Irrigated Lands in Wolf Bay Watershed Area*
Supervisors: Dr. Mark Dougherty and Dr. Latif Kalin
- B.S. Tribhuvan University, Institute of Forestry, Nepal

PUBLICATIONS

44 published, 7 under review/revisions, and 5 first draft ready; *Graduate, Student, ** Undergraduate student, +indicates shared first authorships

Under Review/Revisions

1. Wang M., Shi, W, Bhattarai N, Penny G, Deng X, Ma, R, Qin Y, Xiao, X. Agricultural water sustainability under threat from the water use efficiency paradox. (Under Review)
2. Shayeghi A*, Bhattarai N, Pan L, Jain M, Chen X, Neeson TM, McPherson RA, Penny G, Paudel J, Pradhan N, Shrivastava A, Pegion, K, & Xiao X. Climate-driven Increases in Vapor Pressure Deficit Threaten Winter Wheat Water Productivity in India. (Under Review)

3. Abdoli, N*, Bhattarai N, Paudel J, Xie Y, Koch J, Bedle H, Hydraulic Fracturing is a Key Driver of Land Cover Dynamics in the Permian Basin, United States. (Under Review)
4. Pan B, Xiao X., Bhattarai N, Xie Y, Yin S, Pan L, Yao Y, Celis J, Liu L, Zhang C, Meng C, & Qin Y. Identifying yellow flowers of canola crop with the PlanetScope images and Carotenoids and Yellow Flower Indicator (CYFI). (Under Review)
5. Acharya Sharma B, Maharjan B, Bhattarai N, Das S, Thapa R, Ray R, Hamido S, Zinati G, Fornara D, Panday D, Yang A, Kang S, Kumar R, Wang S. Soil Health and Human Health: A Bibliometric Analysis and Review. (Under Review)
6. Jain, M., Prudente V, Zhou W, Deshpande M, Bhattarai N, Ishtiaque A, Singh, B. Tradeoffs between crop yield, agricultural residue burning, and groundwater depletion in India's wheat belt. [Link]
7. Bhattarai N, Kustas WP, Anderson M, Gao F, et al. A Multi-Sensor, Physically Based Approach to Capture Daily and Abrupt Changes in Vineyard Evapotranspiration. (Under Review)
8. Shrestha R*, Bhattarai N, et al. Evaluation of OpenET, ECOSTRESS JET, and a New Two-source Energy Balance Model products over Winter Wheat Farms in the Southern Great Plains (To be submitted soon to *Agricultural Water Management*)
9. Cruz G, Pokhrel Y, Bhattarai N, Drivers of historical irrigation expansion in Peru and its exposure to Climate Change (To be submitted soon).
10. Shayeghi A*, Bhattarai N, et al., A Satellite-Based Shuttleworth-Wallace Evapotranspiration Modeling Framework for Monitoring Crop Water Use in an Irrigated Desert Olive Orchard in Peru. (To be submitted soon to *Agricultural and Forest Meteorology*).
11. Celis, J, Xiao X, Wagle P, White P, Bhattarai N, & Tubana B. Sugarcane Yield Estimation using Multi-sensor-based High-Resolution Gross Primary Production Modeling. (To be submitted soon to *Agricultural and Forest Meteorology*).

Published

12. Jiang Q, Xu Z, Bhattarai N, Lin Y, Xiao H, Ren J, Pahlow M, Jiang Z, Liu Y, Wu X, Ye G, Zhang H, Li J, Zhu P, Liang S, Cui Y, Liao C, Dong L, Liu J. 2026. Promoting Sustainable Development Worldwide in the Metacoupled Anthropocene. *Nature Communications* 17, 1491. [Link] *Media: [ScienMag](#), [Eureka Alert](#), [Phys.org](#)
13. Whiteis A, Fard S, Suaza-Sierra I, Moreno HA, Bhattarai N, Neeson TM, Long-Term Trends of Striped Bass Habitat Suitability Under a Temperature-Oxygen Squeeze in Lake Texoma, Oklahoma-Texas. *Ecology and Evolution* 16, no. 6: e73736 [Link].
14. Ospina D, Mirazo P, Allan RP, Basnett S, Bastos A, Bhattarai N, with 75 coauthors. 2026. Ten New Insights in Climate Science 2026. *Global Sustainability* 9, e6 [Link].
*Media: [Report and Website](#), [Carbon Brief](#), [World Meteorological Organization \(WMO\)](#), [Stockholm Environment Institute \(SEI\)](#), [Phys.org](#)
15. Wagle P, Shayeghi A*, Bhattarai N, Northup BK, Baral R, Moffet C, Gunter SA, & Baral R. 2026. Assessing Evapotranspiration in Rainfed and Irrigated Alfalfa in the US Southern Great Plains using Eddy Covariance Measurements and OpenET Products. *Agricultural & Forest Meteorology* 379, 111032 [Link].
16. Xu Z, Chen X, Jiang Q, Wu X, Bhattarai N, Mullen J, Gurney G, Li S, Li C, Yin T, Li Z, Fan F, Zhang J, Han S, & Liu J. 2025. Assessing global sustainability performance, imbalance, and coordination over space and time. *Nature Communications* 16, 9186 [Link]. *Media: [SciMag](#), [Bioengineer.org](#)

17. Prudente V, Medina MC, Krishna V, Euler M, [Bhattarai N](#), Lerner A, McDonald AJ, Sherpa S, Rajan H, Urfels A, Santana C, Jain M, 2025. Mapping grain crop sowing date in smallholder systems using optical imagery. *Remote Sensing Applications: Society & Environment* 39:101660 [Link]
18. Jalali J, [Bhattarai N](#), Greene J, Liu T, Marko O, Radulovia M, Sears M, Woznicki SA, 2025. Climate change threatens water resources for major field crops in Serbia's Danube Basin by mid-21st century. *Journal of Hydrology: Regional Studies* 59: 102404 [Link].
19. He L, [Bhattarai N](#), Pokhrel Y, Jia N, Ye G, Song C, Xu Z, Wu S, Li Z. 2024. Dynamics of land cover changes and carbon emissions driven by large dams in China. *iScience* 27:109516 [Link].
20. [Bhattarai N](#), Lobell DB, Singh B, Fishman R, Kustas WP, Pokhrel Y, and Jain M. 2023. Warming temperatures exacerbate groundwater depletion rates in India. *Science Advances*, 35, 9, eadi1401. [Link]. * Media: [NYTimes](#), [The Hindu](#), [The Times of India](#), [The Economic Times](#), [The Weather Channel](#)
21. Hu T, Mallick K, Hitzelberger P, Didry Y, Boulet G, Szantoi Z, Koetz B, Alonso I, Pascolini-Campbell M, Halverson G, Cawse-Nicholson K, Hulley G, Hook S, [Bhattarai N](#), Olivos A, Roujean JL, Gamet P, Su B, 2023. Evaluating European ECOSTRESS Hub Evapotranspiration Products Retrieved from Three Structurally Contrasting SEB Models over Europe. *Water Resources Research*, 59, e2022WR034132 [Link].
22. Zhou Y, Wang H, Liu Z, [Bhattarai N](#), Paudel J, & Qiu H. 2023. Can solar photovoltaic plants reduce carbon emissions & increase income in China? *Environmental Science & Technology*, 57, 49, 20583–20594 [Link]
23. Javed T, [Bhattarai N](#), Acharya BS, Zhang J, 2023. Monitoring agricultural drought in Peshawar Valley, Pakistan using long-term satellite & meteorological data. *Environmental Science and Pollution Research*. [Link]
24. Dhal S, Wyatt B, Mahanta M, [Bhattarai N](#), Sharma S, Rout, T, Saud P, Acharya BS 2023. Internet of Things (IoT) in Digital Agriculture: An Overview. *Agronomy Journal* [Link].
25. Mallick K, Baldocchi D, Jarvis A, Hu T, Trebs I, Sulis M, [Bhattarai N](#), Bossung C, Eid Y, Cleverly J, Beringer J, Woodgate W, Silberstein R, Hinko-Najera N., Meyer WS, Ghent D, Szantoi Z, Boulet G, Kustas WP, 2022. Insights into the Aerodynamic versus Radiometric Surface Temperature Debate in Thermal-based Evaporation Modeling. *Geophysical Research Letters* 49, e2021GL097568 [Link].
26. Bai Y, [Bhattarai N](#), Mallick K, Zhang S, & Zhang J. 2022. Thermally derived evapotranspiration from the Surface Temperature Initiated Closure (STIC) model improves cropland GPP estimates under dry conditions. *Remote Sensing of Environment* 271: 112901 [Link].
27. [Bhattarai N](#), Guido D'Urso, Kustas WB, Bambach-Ortiz N, Knipper KR, Anderson M, Gao F, Alsina, MM, Aboutaleb M, Mckee L, Alfieri JG, McElrone AJ, Prueger JH, & Belfiore OR 2022. Influence of modeling domain & meteorological forcing data on daily evapotranspiration estimates from a Shuttleworth-Wallace model using Sentinel-2 surface reflectance data. *Irrigation Science* 40, 497-513 [Link].
28. Jiang Q, [Bhattarai N](#), Pahlow M, & Xu Z, 2022. Environmental Sustainability and Footprints of Global Aquaculture. *Resources Conservation & Recycling* 180: 106183 [Link].
29. Xu G, Dong H, Xu Z, & [Bhattarai N](#). 2022. China can reach carbon neutrality before 2050 by improving economic development quality. *Energy* 243:12083.[Link]
30. Singh KK, [Bhattarai N](#), & Vukomanovic J. 2022. Landscape-scale hydrologic response of plant invasion relative to native vegetation in urban forests. *Science of the Total Environment*, 802:149903 [Link].
31. [Bhattarai N](#)*, Pollack A**+, Lobell D, Fishman R, Singh B, Dar A, & Jain M. 2021. The impact of groundwater depletion on agricultural production in India. *Environmental Research Letters* 16:085003. [Link]. Media: Mentioned in [BBC News](#)
32. Jain M, Fishman, R, Mondal, P, Galford GL., [Bhattarai N](#), Naeem S, Lall, U, Singh B, & DeFries RS. 2021. Groundwater depletion will reduce cropping intensity in India. *Science Advances* 7: eabd2849 [Link]. Media: [CNN](#), [AAAS](#), [NPR](#), [Earther](#).

33. Trebs I, Mallick K, [Bhattarai N](#), Sulis M, Cleverly J, Woodgate W, Silberstein R., Najera, Hinko-Najera, N., Beringer J, Su Z, & Boulet G. 2021. The role of aerodynamic resistance in thermal remote sensing-based evapotranspiration models. *Remote Sensing of Environment* 264: 112602 [Link].
34. Bai Y, Zhang S, [Bhattarai N](#), Mallick K, Qi L, Tang L, Im J, Guo L, & Zhang J. 2021. On the use of machine learning algorithms to improve cropland evapotranspiration across a wide environmental gradient. *Agricultural and Forest Meteorology*, 288-289: 208308 [Link].
35. Javed T, Zhang J, [Bhattarai N](#), Zhang S, Rashid S, Yun B, Ahmad S, Henchiri M, Kamran M. 2021. Drought characterization across agricultural regions of China using standardized precipitation and vegetation water supply indices. *Journal of Cleaner Production*, 313: 127866 [Link].
36. Rao P, Zhou W, [Bhattarai N](#), Srivastava A, Singh B, Poonia S, Lobell D, and Jain M. 2021. Using Sentinel-1, Sentinel-2, & Planet Imagery to Map Crop Type of Smallholder Farms. *Remote Sensing*, 13: 870 [Link].
37. Khand K, [Bhattarai N](#), Taghvaeian S, Wagle P, Gowda P, & Alderman P. 2021. Modeling evapotranspiration of winter wheat using contextual and pixel-based surface energy balance models. *Transactions of ASABE* [Link].
38. Xu Z, Chen X, Liu J, Zhang Y, Chau S, [Bhattarai N](#), Wang Y, Li Y, Li Y, & Connor T. 2020. Impacts of irrigated agriculture on food–energy water–CO₂ nexus across metacoupled systems. *Nature Communications* 11, 5837. [Link]; 2020 Top 50 Earth, Environmental, and Planetary Sciences Articles; Media: [SciMag](#), [Phys.org](#), [EurekaAlert](#)
39. Niraula R, Saleh A, [Bhattarai N](#), Bajgain R, Kannan N, Osie E, Gowda P, Neel J, Xiao X, & Basara J. 2020. Understanding the effects of pasture type and stocking rate on the hydrology of Southern Great Plains. *Science of the Total Environment*, 708: 134873. [Link]
40. [Bhattarai N](#), Mallick K., Stuart J.**, Vishwakarma, B.D., Niraula, R., Sen, S., & Jain, M. 2019. An automated multi-model evapotranspiration mapping framework using remotely sensed and reanalysis data. *Remote Sensing of Environment*, 229: 69-92. [Link]
41. Cohn A, [Bhattarai N](#), Campolo J, Crompton, O, Dralle D, Duncan J, & Thompson S, 2019. Forest loss in Brazil increases maximum temperatures within 50km. *Environmental Research Letters*, 14: 084047. [Link] Media: [phys.org](#), [Scientific American](#), [Newsroom](#)
42. Kafley H, Lamichane BR, Maharjan R, Khadka M, [Bhattarai N](#), & Gompfer ME, 2019. Tiger and leopard co-occurrence: intraguild interactions in response to human and livestock disturbance, *Basic and Applied Ecology*. [Link]
43. [Bhattarai N](#) & Liu T. 2019. LandMOD ET Mapper: a new Matlab-based graphical user interface (GUI) for automated implementation of SEBAL and METRIC models in thermal imagery. *Environmental Modelling and Software*, 118: 76-82. [Link]
44. Kafley H, Lamichane BR, Maharjan R, Thapaliya B, [Bhattarai N](#), Khadka M, & Gompfer M.E. 2019. Estimating Prey Abundance and Distribution from camera Trap data using binomical mixture models. *European Journal of Wildlife Research* 65: 77. [Link]
45. Mallick K, Wandera L, [Bhattarai N](#), Hostache R, Chormanski J, & Kleniewska M. 2018. A critical evaluation on the role of aerodynamic and canopy-surface conductance parameterization in SEB and SVAT models for simulating evapotranspiration: a case study in the Upper Biebrza National Park wetland. *Water*, 10 (12): 1753. [Link]
46. [Bhattarai N](#), Mallick K, Brunsell NA, Sun G, & Jain M. 2018. Regional evapotranspiration from an image-based implementation of the Surface Temperature Initiated Closure (STIC1.2) model and its validation across an aridity gradient in the conterminous United States, *Hydrology and Earth System Sciences*, 22: 2311-2341. [Link]
47. Niraula R, Meixner T, Dominguez F, [Bhattarai N](#), Rodell M, Ajami H, Gochis D, & Castro C. 2017. How might recharge change under projected climate change in the western US? *Geophysical Research Letters*, 44: 10407-10418. [Link] Media: [UA News](#), [Science Daily](#), [phys.org](#), [AAAS and EurekaAlert](#), [technology.org](#), [futura.org](#)

48. Bhattacharai N, Wagle P, Gowda P, & Kakani V. 2017. Utility of remote sensing-based surface energy balance models to track water stress in rain-fed switchgrass under dry and wet conditions. *ISPRS Journal of Photogrammetry and Remote Sensing*, 133:128-141. [[Link](#)]
49. Richards P, Cohn A, Arima E, VanWey L, & Bhattacharai N, 2017. Enforcement evasion highlights need for independent satellite monitoring for forest governance. *Conservation Letters*, 10:497-498. [[Link](#)]
50. Bhattacharai N, Quackenbush LJ, Im Jungho, & Shaw SB, 2017. A new optimized algorithm for automating endmember pixel selection in the SEBAL and METRIC models. *Remote Sensing of Environment*, 196:178-192. [[Link](#)]
51. Wagle P, Bhattacharai N, Gowda P, & Kakani V. 2017. Performance of five surface energy balance models for estimating daily evapotranspiration in high biomass sorghum. *ISPRS Journal of Photogrammetry and Remote Sensing*, 128:192-203. [[Link](#)]
52. Richards P, Arima E, VanWey L, Cohn A, & Bhattacharai N 2017. Are Brazil's Deforesters Avoiding Detection? *Conservation Letters*, 10:470-476. [[Link](#)] Media: [Nature Climate Change](#); [Mongabay](#), [phys.org](#), [eurekalert](#), [Brown University](#), [Nature World News](#), [Technology.org](#)
53. Bhattacharai N, Shaw SB, Quackenbush LJ, Im J, & Niraula R. 2016. Evaluating five remote sensing-based single-source surface energy balance models for estimating daily evapotranspiration rates in a humid subtropical climate. *International Journal of Applied Earth Observation and Geoinformation*, 49:75-86 [[Link](#)]
54. Bhattacharai N, Quackenbush LJ, Dougherty M, & Marzen L, 2015. A simple Landsat-MODIS fusion approach for monitoring seasonal evapotranspiration at 30 m spatial resolution. *International Journal of Remote Sensing*, 36:115-143. [[Link](#)]
55. Shaw SB, Marrs J**, Bhattacharai N, & Quackenbush LJ, 2014. Longitudinal Study of the Impacts of Land Cover Change on Hydrologic Response in Four Mesoscale Watersheds in New York State, USA. *Journal of Hydrology*, 519:12-22. [[Link](#)]
56. Bhattacharai N, Dougherty M, Marzen L, & Kalin L. 2012. Validation of evaporation estimates from a modified surface energy balance algorithm for land model in the south-eastern US. *Remote sensing letters*, 3:511-519. [[Link](#)]

Book Chapter

1. Bhattacharai, N, Shayeghi, A*, & Jain, M, 2025. Spatiotemporal distribution and recent trends in crop water use across India (Invited). In *Remote Sensing of Land Cover and Land Use Changes in South and Southeast Asia*, Volume 2 (pp. 56-67). CRC Press. Editors: Krishna P. Vadrevu, Chris Justice, & Garik Gutman.

Research grants and contributed proposals

Funded Projects

1. **USDA NIFA** ($\approx$ \$300K). **PI:** Nishan Bhattacharai; Co-PIs: Jayash Paudel (OU) and Sean A. Woznicki (Grand Valley State University), 2026-2028.
2. **NSF E-RISE: Center for Grassland Ecosystem Forecasting** ($\approx$ \$8 million). My role: co-PI with PI Xiangming Xiao and faculty members at OU, 2026-2030
3. **UNSA Global Change and Human Health (GCHH) Institute** ($\approx$ \$4.2 million). My role: co-PI (Nishan Bhattacharai) with several faculty members at OU and UNSA/**PI:** Tim Filley. Period: 2025-2027.
4. **NASA EPSCoR** ($\approx$ \$30K). **PI:** Nishan Bhattacharai; Collaborator: Simon J Hook (NASA JPL). Period: Jan-May 2026.
5. **USDA-ARS** ($\approx$ \$9K). **PI:** Nishan Bhattacharai; Collaborator: Pradeep Wagle. Period: May 2026- April 2027.
6. **USGS South Central Climate Adaptation Center** ($\approx$ \$71K). **PI:** Nishan Bhattacharai, Period: 2024-2026.
7. **NSF STTR Phase I** (\$275K). My role: co-PI (Nishan Bhattacharai)/**PI:** Jorge Celis, Period: 2024-2025

8. **USGS 104b Water Center** (~\$29K). **PI:** Nishan Bhattarai. Period: Sep 2025-July 2026.
9. **Big Idea Challenge, VPRP University of Oklahoma** (~\$300K). My role: co-PI Nishan Bhattarai/**PI:** Kathy Pegen. Period: 2024-2026.
10. **Data Institute for Societal Challenges** (~\$10K and \$13K), **PI:** Nishan Bhattarai, Co-I: Pradeep Wagle. Period: Summer 2023 and 2026-2027.
11. **NASA OK EPSCoR** (~\$2K). **PI:** Nishan Bhattarai, Collaborator: Simon J Hook (NASA JPL). Period: Jan-May 2025.
12. **NASA:** NNH19ZDA001N-LCLUC (~\$448K; \$10K to OU). My role: Collaborator (Nishan Bhattarai)/PI Meha Jain. **Period:** Jan 2021- Dec 2024.
13. **NASA** NNH21ZDA001N-LCLUC (~\$450K). My role: Collaborator (Nishan Bhattarai)/**PI:** Sean Woznicki (Grand Valley State University), Period: Jan 2022- Dec 2025.
14. **NCAR** Computational & Information Systems Laboratory. Cheyenne (SGI ICE XA Cluster) (100,000.0 Core-hours). Project # UMIC0046, 2016- 2019. **PI:** Nishan Bhattarai
15. **Raymond Von Dran Fund** (\$2K). Project: Micro-Hydro consultants, Syracuse University. **PIs:** Nishan Bhattarai, John McDonald, and Prakhya Thapa. Period: Summer 2012.

TEACHING EXPERIENCE

University of Oklahoma (Teaching 4-5 courses a year)

GIS 4133/5133 *Fundamentals of Remote Sensing* (Fall 2024, Fall 2022, Fall 2023, Spring 2024, Spring 2025, Fall 2025, Spring 2026, Fall 2026) – 3 credits

GIS 5133 (online) *Fundamentals of Remote Sensing* (Spring 2023, Summer 2023, Spring 2024, Summer 2024, Spring 2025, Spring 2026, Summer 2026) – 3 credits

GEOG 4293/5293 *Hydrologic Science* (Fall 2023, Fall 2024, Fall 2025) – 3 credits

GIS 6240 *Global Water Sustainability Challenges* (3 credits, Spring 2025)

University of Michigan

Co-Instructor, EAS 501 *Introduction to R (Natural Resources Statistics)*, 1 credit (Co-designed and taught Introduction to R programming for beginners, Spring 2019)

Guest Lecturer, EAS 501 *Watershed Hydrology & Water Resources Management* (Spring 2020)

SUNY-ESF

Teaching Assistant (Aug 2011- 2014): Courses: *GIS for engineers* (Fall 2011-2013); *Statics & Dynamics* (Spring 2012); *Mechanics of Materials* (Spring 2012); *Introduction to Engineering Design* (Spring 2012-2014)

Auburn University

Teaching Assistant (2009)-*Introduction to renewable resources*

STUDENT ADVISING AND MENTORSHIP

Graduate Students (Advisees):

Afshin Shayeghi (Ph.D. Geography & Environmental Sustainability, GES, Fall 2023-),

Nastaran Abdoli (PhD, Geography & Environmental Sustainability, Fall 2024-, PhD. Candidate),

Graduated (n= 3)

Aiyasha Ghani (MS 2025; Currently ORISE Fellow at US Forest Service)

Aarati Kafle (MS, 2026)

Reeya Shrestha (MS 2026)

Graduate Committees: Shadi Fathollahifard (Ph.D. GES, OU, Hananeh Omid (Ph.D., GES, OU), Jamshid Jalali (MS 2025, Grand Valley State University), Baihong Pan (Ph.D. Biology, OU), Yuan

CV- Nishan Bhattarai (June 2026) - 6

Yao (Ph.D. Biology, OU), Ali Shojaeian (Ph.D., Civil Engineering, OU), Sadiksha Rai (Ph.D., Meteorology, OU), Li Pan (Ph.D. Biology 2025, OU), Aman Bhatta (Ph.D., Environment and Sustainability, University of Michigan), Jorge Andrés Celis (Ph.D. 2023, OU), Kritika Pathak (MS 2023, OU), Akpoezi R. Ononeme (MS 2023, GES, OU), Ming Zhu (Ph.D., Meteorology, OU)

Graduated (n = 10): Baihong Pan (Ph.D. OU, 2026), Himel Tadulker (Ph.D. OU, 2026), Jorge Andrés Celis (Ph.D. 2023, OU), Li Pan (Ph.D. Biology, OU, 2025), Ally Whiteis (MS 2026, OU), Kritika Pathak (MS 2023, OU), Akpoezi R. Ononeme (MS 2023, OU); Jamshid Jalali (MS 2025, Grand Valley State University), Nikhyal R. Poudyal (MS 2024, OU), Ali Shojaeian (Ph.D. Civil Engineering 2025, OU)

Undergraduate students Mentored

University of Oklahoma: Peter Mueller (2026), Xochitl Hidalgo (2024-2025), Kaleigh Dennis (2024-2025), Richard Garcia (Spring 2023), Ryan Penic (Spring 2023), Tim White (Spring 2023), Sujal Shrestha (Fall 2023)

University of Michigan: Shon Harris, Julia Stuart, & Reese Jia Er Siew

SUNY-ESF: Andrew Sussman, John Marrs, and Prakhyat Thapa

Other mentorship experience: *STEM Mentor/Elementary school teacher* (Jan 2015 – May 2015)

AWARDS AND HONORS

- Group Award for Excellence in Transdisciplinary, Convergent Research, OU VPRP (2025)
- Research Achievement Award, USDA-ARS (2021)
- Outstanding reviewer (2018), *Journal of Hydrological Engineering (ASCE)*
- ERE Departmental Award for Academic Excellence (2014), SUNY-ESF
- AGU Student Travel Award (2013), American Geophysical Union (AGU), \$500
- CNY Graduate Student of the Year (2013), ASPRS, \$500
- Ta Liang Memorial Award (2013), ASPRS, \$2,000
- Research In Need Grant (\$250), SUNY-ESF, Summer 2012
- SUNY-ESF graduate student travel Awards (\$250 and \$500), SUNY-ESF, 2012-2013
- ConForM/Danida fellowship (~\$250) for B.S. thesis work (2005-2006)
- Merit Scholarship, Tribhuvan University (2002- 2006)
- BSc Entrance Topper (science stream), Tribhuvan University (2002)

Awards won by students

- Afshin Shayeghi (Ta Liang Memorial Award, \$3000, ASPRS)
- Nastaran Abdoli and Afshin Shayeghi (Robberson Travel Award, \$1000, OU)
- Afshin Shayeghi (Standley Award, Outstanding publication by a graduate student, \$1000, OU)
- Afshin Shayeghi (First Prize, \$1000), GIS Day poster, OU)
- Afshin Shayeghi (Graduate Student Senate Grant, \$750, OU)
- Sujal Shrestha (Undergraduate Research Award, \$900, OU)
- Julia Stuart, 2018 South Asia Fellow (~\$3000), International Institute at the University of Michigan
- Julia Stuart, 2021, Hugh G. Rumler Prize (outstanding senior), University of Michigan

SELECTED CONFERENCE PROCEEDINGS/ABSTRACTS/EDITORIAL

1. Kustas, W.P., Knipper, K., et al. with Bhattacharai, N. 2024. Observations from Leaf and Canopy to Field, Landscape and Regional Scales for Understanding Soil-Plant-Atmosphere Exchange Processes Influencing Evapotranspiration in Vineyards. American Meteorological Society, Baltimore, MD [Link]
2. Mallick K Baldocchi, et al. with Bhattacharai N 2023. Consort of Conductances: The Missing Biophysical Link in Thermal Remote Sensing of Terrestrial Evaporation and Inclusion for the Future LST Missions. European Space Agency- ESRIN, Italy.

3. Reddy PK, Bhattarai N, & Sen S, 2023. Understanding Evapotranspiration Variability between the Eastern and Western Himalayas. EGU General Assembly 2023 [Link]
4. Belfiore OR, Kustas WP, et al. with Bhattarai N 2023. Estimating ET by using canopy conductance models with Sentinel-2 data in irrigated crops in California and Australia. A European vision for hydrological observations and experimentation, Naples, Italy, 12–15 Jun 2023, GC8-Hydro-58 [Link].
5. Bhattarai N & Wagle P, Recent advances in remote sensing of evapotranspiration. *Remote Sensing*. 2021; 13(21):4260. [link]
6. Bhattarai N, Kustas WP, et al. 2021. Synergistic use of spectral information from Landsat & Sentinel-2 data for modeling near real-time crop water status across California vineyards. AGU Fall meetings 2021.
7. Mallick K, Hu T, Yun B, Bhattarai N, et al. Thermal and shortwave infrared remote sensing of ecosystem processes: Opportunities, synergies, and challenges. IEEE InGARSS 2021, 440-443 [Link]
8. Kustas WP, Bhattarai N, et al. Daily ET estimates from application of Shuttleworth-Wallace model with Sentinel-2 surface reflectance data over California vineyards. AGU Fall Meetings 2021 [Link]
9. Bhattarai N, Mallick K, & Jain M, 2019. Sensitivity of four contextual remote sensing-based SEB models to spatial domain, *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, XLII-3/W6, 3-7 (Peer-reviewed) [Link]
10. Bhattarai N, Quackenbush LJ, Im J, & Shaw, SB, Automation of Endmember Pixel Selection in SEBAL/METRIC Model. AGU Fall Meetings Abstracts, December 14-18, 2015, San Francisco, CA.
11. Bhattarai N, Quackenbush LJ, & Shaw, SB 2014. Comparison of four single-source surface energy balance-based models for estimating remotely sensed daily ET. Abstracts from the ASABE 2014 International Symposium on ET. April 7-11, 2014, Raleigh, NC.
12. Bhattarai N & Quackenbush LJ 2013. A data fusion approach for monitoring remotely sensed seasonal ET. AGU Fall Meetings Abstracts, December 9-13, 2013, San Francisco, CA.
13. Bhattarai N, Quackenbush LJ, Calandra L, Im J, & Teale S, 2012. An automated object-based approach to detect Sirex-infestation in pines. Proceedings of American Society for Photogrammetry and Remote Sensing (ASPRS) Annual Conference, March 19-23, 2012, Sacramento, CA.
14. Bhattarai N, Quackenbush LJ, Calandra L, Im J, & Teale S, 2011. Spectral analysis of Scotch pine infested by Sirex Noctillo. Proceedings of ASPRS Annual Conference, May 1-5, 2011, Milwaukee, WI.

INVITED TALKS

US CLIVAR (2026), NASA (2024), Tarleton State University (2024), University of Oklahoma (2022), Ohio State University (2022), University of Houston (2022), Lawrence Berkeley National Lab (2020), College of William and Mary (2019), Western Michigan University (2020).

CONFERENCE PRESENTATIONS

1. A multi-sensor and multi-model synergy to enhance real-time monitoring of evapotranspiration at 10m spatial resolution to support sustainable irrigation water management. AGU Fall Meetings 2023.
2. Assessing the utility of evapotranspiration products in monitoring agricultural responses to changing climate (Poster presented by Ph.D. student Afshin Shayeghi at the AGU Fall meetings 2023)
3. A Deep Learning-Based Ensemble Surface Energy Balance Modeling Approach to Monitor Crop Water Use & Water Stress in drylands, AGU Frontiers in Hydrology, June 20, 2022, San Juan, PR

4. Synergistic use of spectral information from Landsat and Sentinel-2 data for modeling near real-time crop water status across California vineyards, December 15, 2021, AGU Fall Meetings, New Orleans, LA
5. Warming Temperatures Lead to Increased Groundwater Depletion in India, Dec 11, 2019, AGU Fall Meetings, San Francisco, CA.
6. Understanding the impacts of groundwater depletion and climate shocks on irrigation decisions in India. AGU Fall Meeting, Dec 10-14, 2018, Washington, DC.
7. An automated multi-model based evapotranspiration estimation framework for understanding crop-climate interactions in India. AGU Fall Meeting, Dec 11-15, 2017, New Orleans, LA.
8. Understanding the climate-included variations in the seasonal water demands of irrigated crops in Northern India. AGU Fall Meeting, Dec 11-16, 2016, San Francisco, CA.
9. Introduction of automated calibration approaches to the surface energy balance-based ET algorithm, ASPRS Annual Conference, Mar 23-27, 2014, Louisville, KY.
10. Comparison of four single-source surface energy balance-based models for estimating remotely sensed daily ET. ASABE 2014 International Symposium on ET. Apr 7-11, 2014, Raleigh, NC.
11. Application of remote sensing and surface energy balance algorithms in estimating ET in the southeastern US. 24th ASPRS 2013 annual conference, Mar 24-28, 2013, Baltimore, MD.
12. Using remote sensing and geospatial techniques in hydrological applications. NYGeoCon. NYGIS Association, Nov12-13, 2013, Saratoga Springs, NY.
13. Calibration of the InVEST water yield model- An automated approach, World Wildlife Fund-US, Aug 9, 2013, Washington, DC.
14. A coupled multi-sensor fusion & SEB algorithm approach to derive spatially-distributed seasonal ET. 22nd GIS/SIG Annual Spatial/Digital Mapping Conference, Apr 2013, NY.
15. An automated object-based approach to detect Sirex-infestation in pines. 23rd ASPRS 2012 annual conference, Mar 19-23, 2012, Sacramento, CA.

FIELD/SUMMER EXPERIENCE

- *Research/Field Tech* (SapFlux & weather station), University of Illinois (Jun - Aug 2014)
- *Conservation Science Intern*, World Wildlife Fund for Nature, Washington, DC (Jun - Aug 2013)
- *Research Aide*, SUNY-ESF (Jun -Aug 2012)
- *International Corps Member*, EarthCorps, Seattle, WA (Jun-Dec 2007)

PROFESSIONAL SERVICES

University of Oklahoma

- Graduate Committee, DGES
- Faculty Search Committee Member (DGES, School of Meteorology, and CEES)
- Scheduling Committee, DGES
- Faculty Appeals Committee

Editor: Guest Editor, Special Issue on Artificial Intelligence in Coastal Wetlands Remote Sensing. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*. (December 1, 2025 - Present).

Editorial Board member of *GIScience & Remote Sensing* (Taylor & Francis)

Guest Editor for *Remote Sensing's* Special Issue on Remote Sensing of Evapotranspiration (2020)

Peer reviewer (~170 articles for over 50+ journals): *Nature Food*; *Nature Communications*; *PNAS*; *Remote Sensing of Environment*; *Global Change Biology*; *PNAS Nexus*; *Environmental Research Letters*; *Geophysical Research Letters*; *Journal of Geophysical Research-Atmospheres*; *Water Resources Research*; *Journal of Hydrology*; *Journal of Hydrometeorology*; *Earth's Future*; *ISPRS Journal of Photogrammetry & Remote Sensing*; *IEEE TGRS*; *IEEE Geoscience & Remote Sensing Letters*; *IEEE-JSTARS*; *International Journal of Applied Earth Observation & Geo-*

information; Journal of Hydrologic Engineering; Scientific Reports; Agricultural & Forest Meteorology; International Journal of Remote Sensing; Science of the Total Environment; Hydrological Processes; Remote Sensing; Stochastic Environmental Research & Risk Assessment; GIScience & Remote Sensing; PLoS ONE; Hydrology; Water; Transactions of the ASABE; Applied Water Science; Sustainability; Sensors; Remote Sensing Letters; Land Use Policy; Computers & Geosciences; Environmental Processes; Irrigation & Drainage Engineering; Hydrological Sciences Journal; Journal of Forestry & Natural Resource Management; Environmental Modelling & Software; Advances in Space Research; Agricultural Water Management; International Journal of Commons.

Proposal Reviewer

- Served in NASA proposal (2020, 2023, and 2025) review panels
- NASA postdoctoral program proposal reviewer (2023-2026)
- NSF proposal reviewer (2024, 2026)
- BELSPO, Belgian Earth Observation STEREO IV programme
- External Project Advisor for International Initiative for Impact Evaluation (3ie; 2016-2018)
- Proposal reviewer for the Citrus Research Board (2021).

Scholarship panel: American Society for Photogrammetry & Remote Sensing Award Committee (ASPRS, 2022-).

Professional Memberships: American Geophysical Union (AGU, 2013-), American Society for Photogrammetry and Remote Sensing (ASPRS, 2010-2017, 2023), European Geophysical Union (EGU, 2018), American Society of Agricultural & Biological Engineers (2014).

COMPUTER SKILLS

GitHub Page

- Fluent in multiple programming platforms, including R, Matlab, Python, & Google Earth Engine (GEE)
- Experienced user of High-Performance Computing (HPCs)/Clusters and Stata
- GIS/Remote Sensing software, such as ArcGIS, QGIS, ENVI, ERDAS Imagine
- Basic knowledge of C/C++ and Visual Basic
- knowledge of other software, such as AutoCAD, HEC-HMS, SAS, SigmaPlot, SSH, Unix shell scripts, Git, and SQL Server.